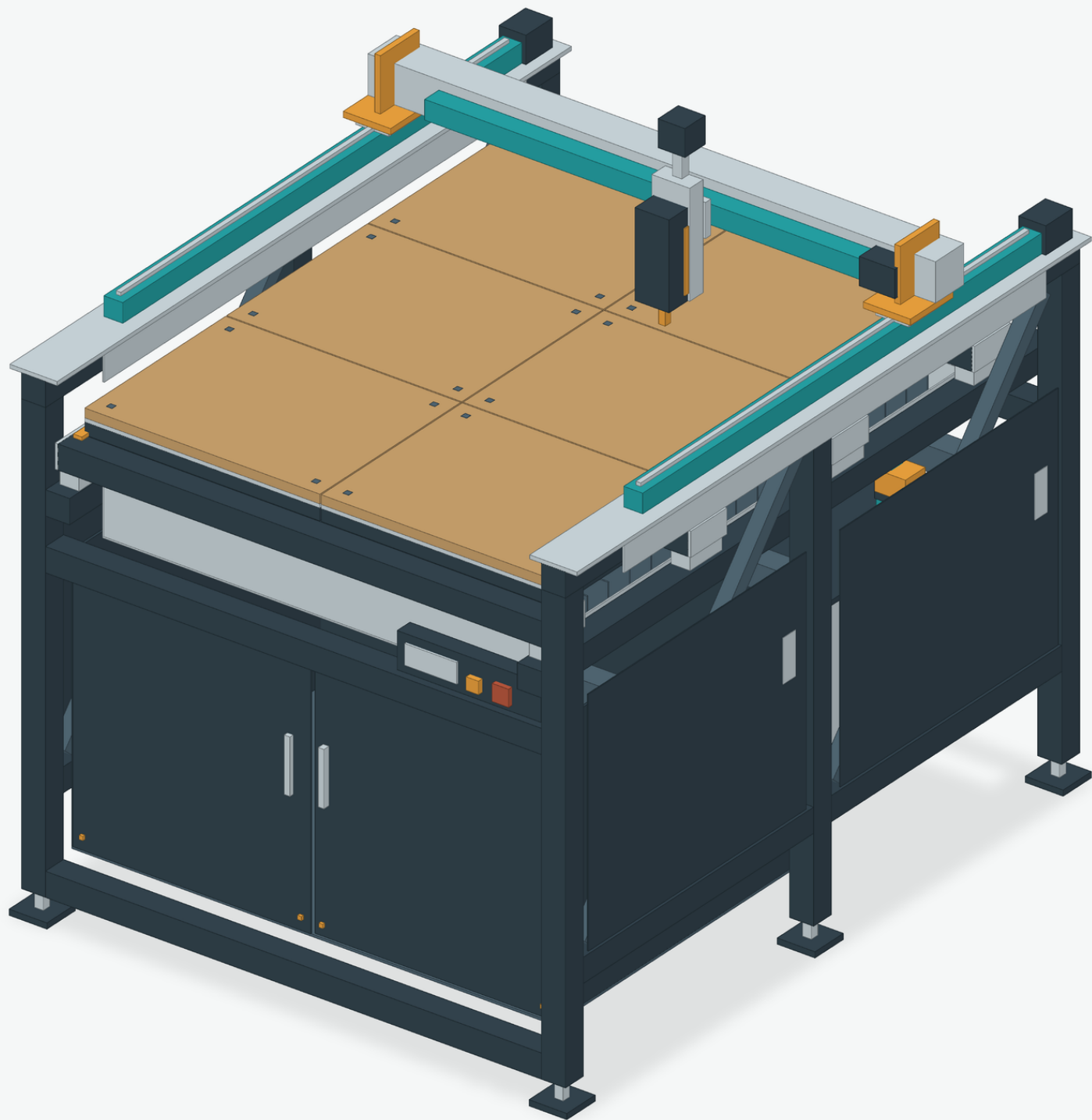


The router bed now has a real docking system.

Fixed wet pan. Four removable bridge beams. Six rigid deck panels. Every part stores inside the chassis.



Chassis: 1150 W x 1450 D. No swap cart or external bed-storage station.

01 / INTEGRATED BED SYSTEM

One finished machine

A structural bridge over the pan

Paired steel beams land on metal receiver seats tied directly to all six chassis legs. No routing load passes through the water tray or slats.

Six numbered, rigid deck panels

Each panel has a welded steel frame, an 8 mm aluminum subplate and replaceable MDF. Pins locate it; metal-to-metal clamps hold it down.

Guided drain and refill

Fixed plumbing sends water into a vented settling reservoir. No hose disconnection or wet cassette lift is needed for a normal mode change.

Everything has a parking place

Lightweight lift-off covers reveal the full-width cabinet. Covers park on side studs within the rail overhang; angled beam slots and numbered panel banks keep the changeover orderly.

A deliberate exterior

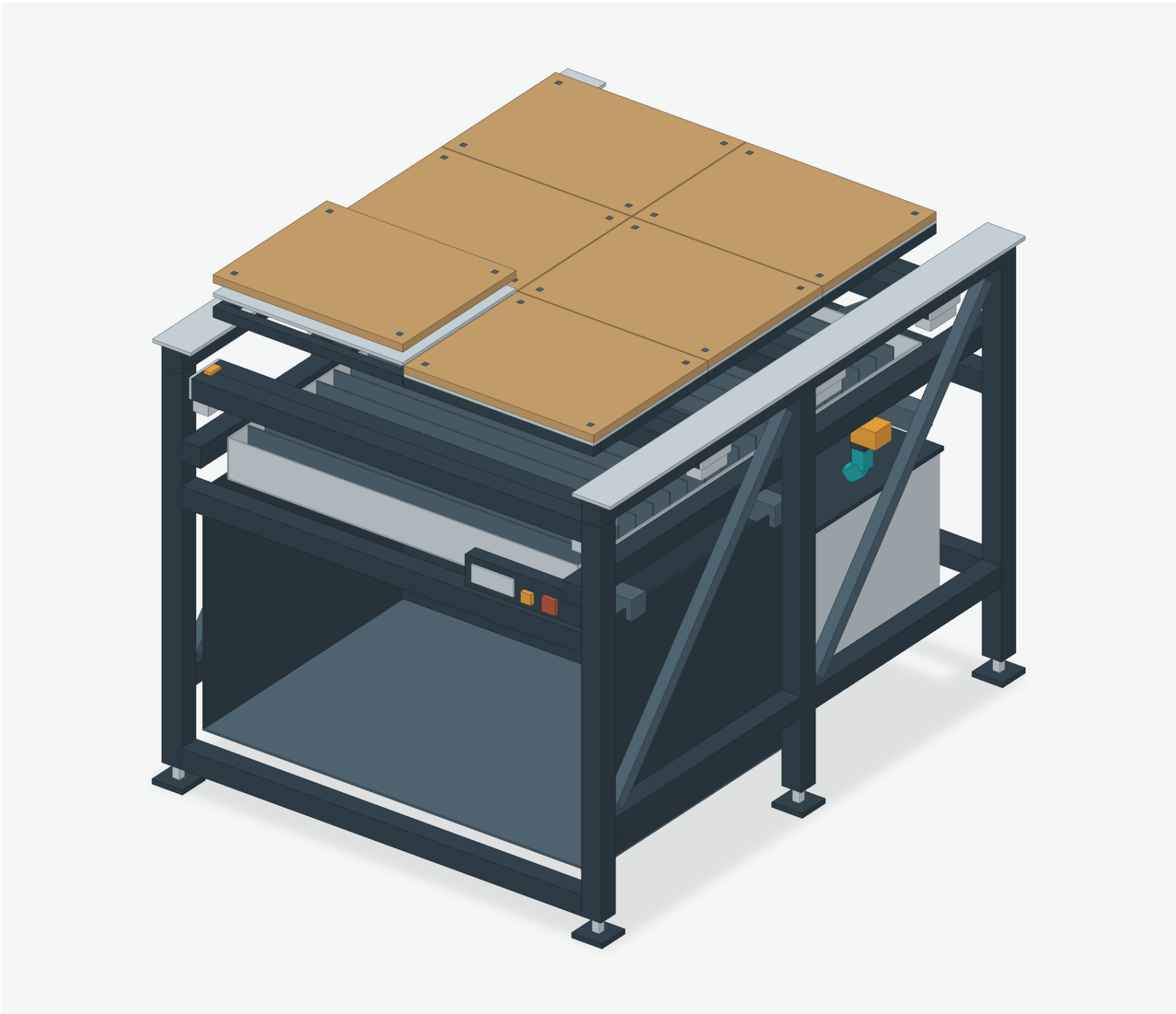
Graphite skins conceal the braces; removable brushed-metal shields protect the wet zone. Amber marks identify release points and clamps. Service panels remain accessible.

REVISION C REPLACES REVISION B

The rail-support height increases from 930 to 1050. Plasma stays at 850; the new router deck starts at 972.8 before surfacing. Separate head mounting references accommodate the two planes.

Exactly how the CNC bed fits

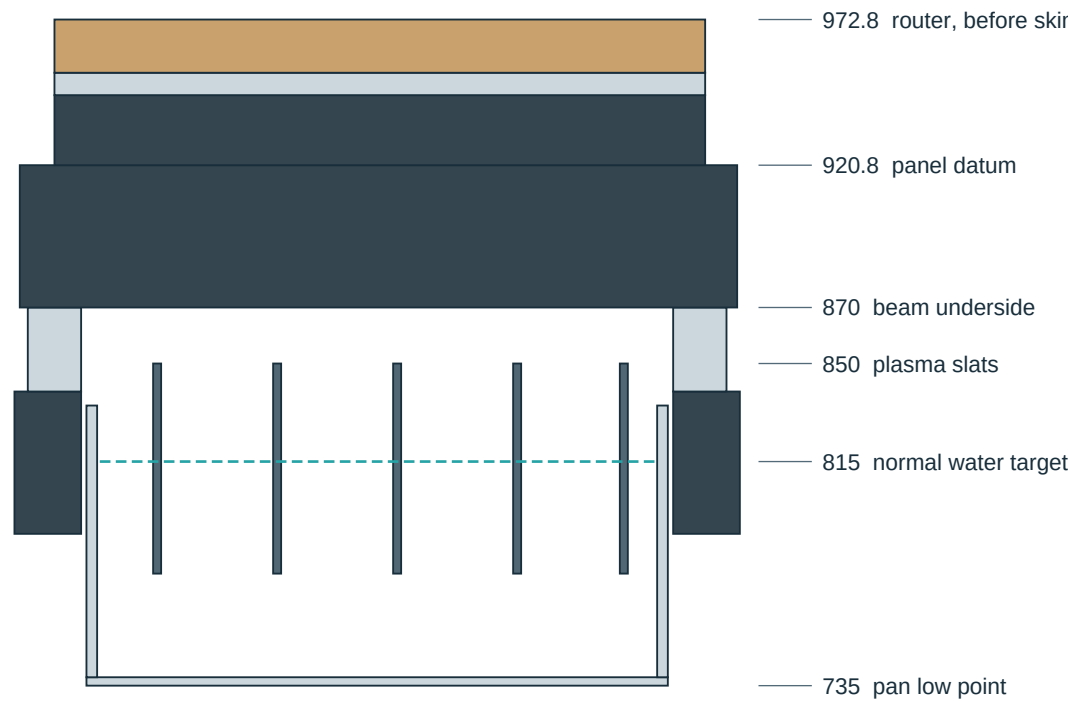
The removable deck bridges the fixed pan. Its supports stay clear of plasma sheet loading when removed.



EXPLODED ASSEMBLY - vertical separation is illustrative; gantry hidden.

02 / SIDE SECTION

Two work planes



20 mm nominal slat-to-beam clearance

Fixed ledgers stop at 840, below the slats. The removable beams carry their own 30 mm riser feet, so their undersides sit at 870. Verify clearance to warped slats before installing the deck.

LOAD PATH: cutter -> MDF/aluminum panel -> steel panel frame -> paired bridge beam -> hard receiver seats -> six-leg chassis -> floor.

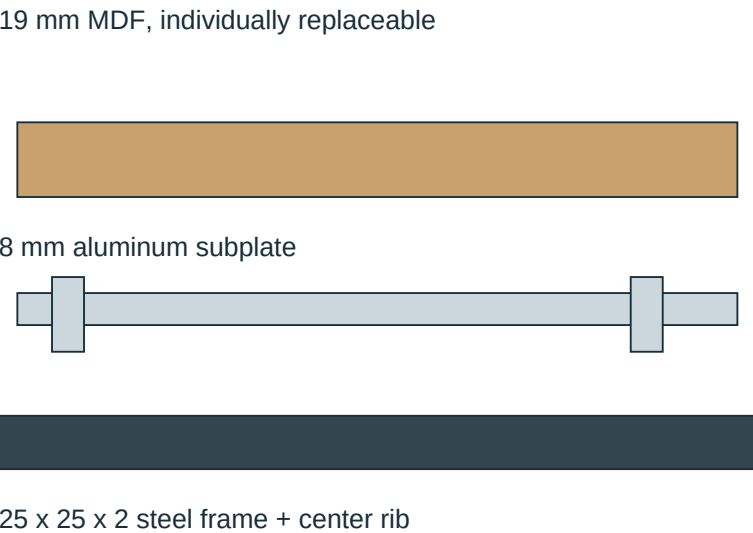
The 100 mm Z stroke does not span both work planes by itself. The torch and router need independently located mounting references, set from their actual tool-tip geometry.

Rigid contact, repeatable location, captive hardware

Precision is established at the metal contacts; the MDF is replaceable and is surfaced after the complete deck is installed.

A / ONE DECK PANEL

Small enough to handle



Actual panel blank: 497 x 397. Six panels form a 997 x 1197 deck with 3 mm seams. Each completed panel is approximately 10-11 kg. Handles fold flat; numbered corner keys prevent transposition.

B / LOCATORS AND CLAMPS

Locate first. Pull down second.

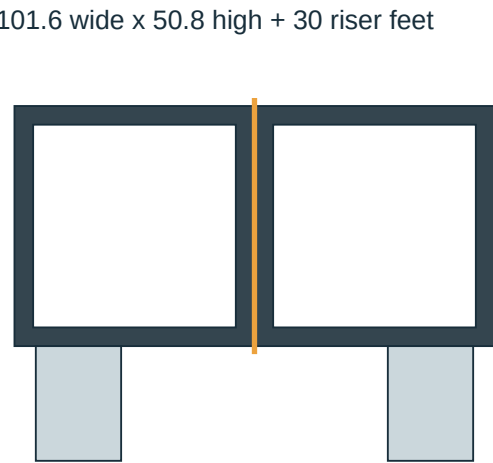


Use hardened mating inserts, a round locator and a relieved diamond locator. Captive M8 draw-down screws load metal sleeves directly above supports; MDF is not in the clamping stack. Recessed access plugs cover the screw pockets.

Clean and inspect the contact pads at each swap. Keep paint, rubber and chips off the datum faces. Final pin fits and screw engagement follow the selected hardware drawings.

C / REMOVABLE CROSSBEAM

A tied pair of 2-inch tubes



1044 finished overall; 1032 tube blanks

Two tubes are connected along their meeting seams and at load-transfer points, with 6 mm end diaphragms. Ground or shimmed contact pads set the datum after welding. End plates alone do not establish equal load sharing.

Beam clamps pull the riser feet onto fixed steel seats. Locators carry lateral forces. Clamp engagement sensors support the changeover sequence; they do not prove clamp preload.

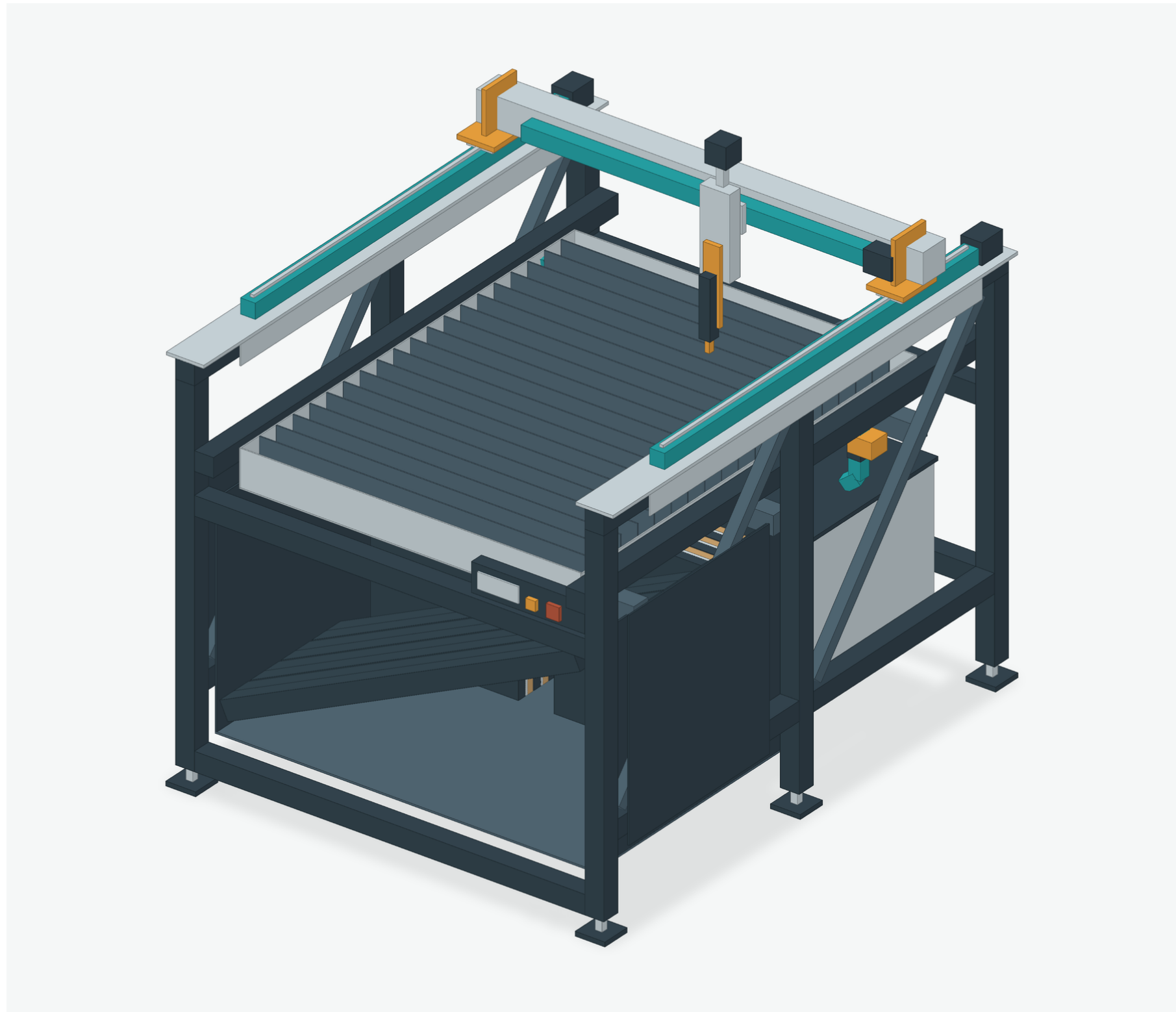
STIFFNESS CHECK / ILLUSTRATIVE, NOT A TEST RESULT

A tied pair of 2 x 2 x 0.120 tubes, simply supported over about 1 m, deflects approximately 0.023 mm under a 100 N central load if both tubes share that load. One tube carrying it alone gives about 0.047 mm.

This calculation excludes joints, panels, gantry and axes. Commission the assembled bed with a known force and an indicator, then repeat at seams and corners. Suggested starting targets: ≤ 0.10 mm bed movement at 100 N and ≤ 0.05 mm height change after a clean re-dock; verify these experimentally.

A swap that stays inside the machine footprint

No full-bed lift, no moving water pan and no long drawer. The storage order matches the assembly order.



CUTAWAY / skins and doors removed only to show storage

Storage allowance: approximately 1005 W x 550 H x 680 D clear, with the full opening exposed. Beam assemblies park around 25 degrees; two banks of three panels stand behind them. Verify feet, clamps, handles and cover-parking clearance in the mock-up.

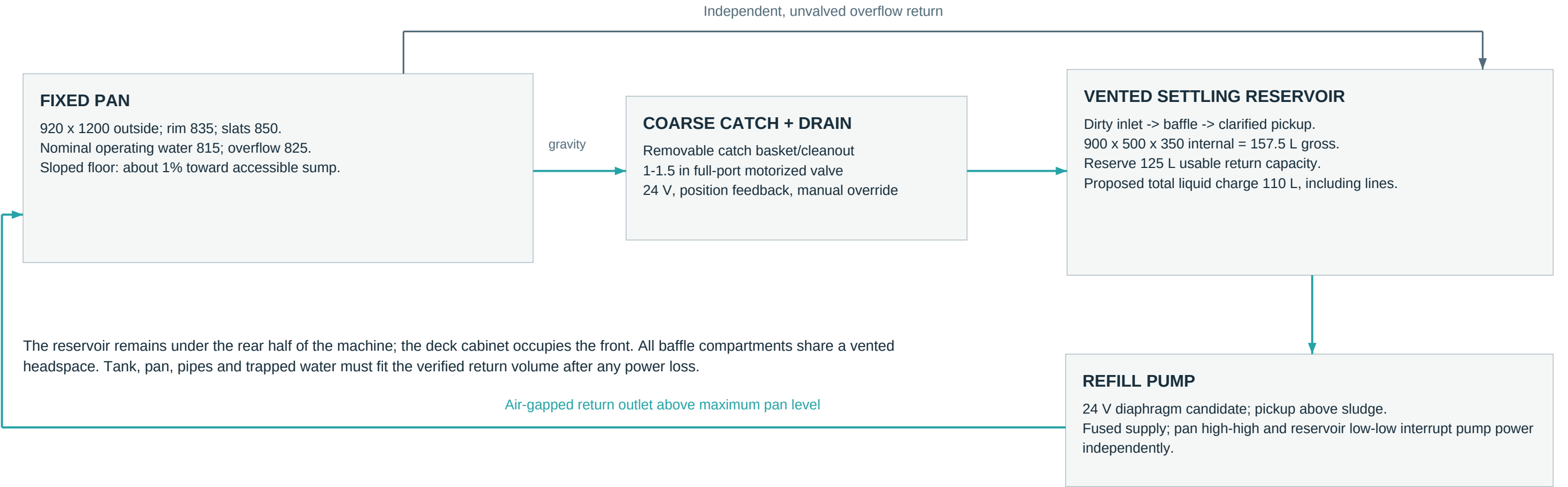
03 / ROUTER CHANGEOVER

A guided sequence

- 1 Isolate and remove the torch head**
Finish the job and inhibit machining. Remove the torch head cassette; park the bare Z fully up at the rear. Torch retraction alone may not clear the higher deck.
- 2 Run the drain cycle**
Verify return capacity, open the drain, monitor low level, then close it. The panel reports drained to low level, not dry. Inspect and clean residual water.
- 3 Dock the four beams**
Lift off the light front covers and park them on the stand. Remove beams from their angled slots; seat riser feet, engage locators and tighten captive beam clamps.
- 4 Fit the six panels**
The panel banks are now accessible. Fit numbered panels to their keyed locations; clean contacts and tighten the metal draw-downs.
- 5 Fit the router head and probe**
With the deck locked, fit the router head at its indexed reference. Connect dust collection, probe the bed and check the tool sweep. New or resurfaced MDF needs a fresh datum.
- 6 Return to plasma in reverse**
Stop and remove the router head first. Stow panels, then beams. Clean wood dust; confirm the dry deck is absent before refill. Fit and identify the torch head before enabling plasma.

Automatic water handling, with a fixed wet system

Normal changes retain every hose connection. Valve, pump, sensors and cleanout are accessible from the rear wet-service bay.



Drain permission	Drain completion	Refill permission	Fault and restart
Dedicated changeover only. Machining inhibited; pump off; wet system present; reservoir installed with sufficient capacity. Valve-open feedback and a separate drain timeout are required.	Stable pan-low signal, then close valve and verify closed feedback. Inspect puddles and sludge. An actuator end switch proves position, not a leak-tight valve seat or a dry pan.	All six panels and four beams absent; dry dust removed; drain closed; overflow clear. Stop at normal level. Pan high-high and reservoir low-low also stop the pump through an independent circuit.	Timeout, no level progress, inconsistent sensors or reservoir high-high latches a fault. Pump stays off after power loss; reconcile levels and valve position before operator restart. No automatic machining start.

Dirty-water component selection remains open: coarse screening does not remove abrasive fines. The candidate motorized valve and refill pump require confirmation for the actual settled water and additives. Keep the valve, basket, baffles and level switches removable for cleaning. Commission blocked-drain, overflow, independent-trip and interrupted-power cases; measure cycle times instead of promising a drain time.

Revision C material basis and completion checks

This replaces the earlier bed concept and 930 mm rail-support height. Exact motor/head interfaces still require their supplier drawings.

PRELIMINARY FABRICATION BASIS

Qty	Size / length	Assembly	Material / note
2	1450	Top Y-support tubes	2 x 2 x 0.120 steel
6	949.2	Legs	50 foot allowance; steel top 1050
4	648.8	Lower side ties	2 x 2 x 0.120 steel
4	1048.4	End ties	Front lower tie top 100.8
2	1450	Fixed receiver ledgers	2 x 2 tube; top 840
3	1032.4 nominal	Fixed pan bearers	2 x 2 + 8 end plates; fit clearance
8	1032	Removable beam tube blanks	Pairs; 6 end plates -> 1044 finished
8	30 rise	Removable beam feet	Remain with beams; pad faces set datum
6	497 x 397 x 52	Complete panel modules	25 steel frame + 8 AI + 19 MDF
6 sets	25 x 25 x 2	Panel steel frames	2 x 497 + 3 x 347 per panel
1	920 x 1200 x 100	Fixed pan	Floor 735 low point; slats 850
1	900 x 500 x 350 inside	Vented reservoir	157.5 L gross; wall/support extra
1	1005 x 550 x 680	Cabinet clear space	Lift-off covers; diagonal beam rack
1	1200	8080 gantry	Exact section/mass verification pending

Stationary tube allowance: 18.284 m, approximately 84 kg steel. Dry sand: about 36.5 L / 58 kg at assumed 1.6 kg/L. Braces, caps, pan bearers, skins, beds, feet and stock allowance are additional. Leave sand out of the removable deck parts.

PRIMARY REFERENCES / CLICK TO OPEN

Your HMS40 X; matching 1000 mm Y uses B0C7GN24S1

Your RATMMOTOR ZBX80 100 mm Z

Carr Lane locating-pin principles

Valworx 24 V valve example - confirm media suitability

Whale Gulper 320 maker guide - refill candidate only

SJE low-current water/wastewater level switches

WHAT STILL NEEDS TO BE PROVED

Whole-machine mass

Stored deck parts and reservoir water stay on the stand. The revised machine can approach roughly 450-550 kg before heavy stock, depending on skins and hardware. Size feet, ledgers, welds, tank shelf and floor supports for the complete load; do not count only the active bed.

Head geometry and clearance

Plasma is 850; fresh router deck 972.8. Bridge underside 1127 gives 154.2 nominal clearance above the bare router bed, not usable material thickness. Verify the exact spindle, torch, dust shoe and 100 mm Z with their separate located plates.

Motion-system capacity

Retain the chosen HMS40 modules, but check the assembled head and bridge masses, carriage moments and operating speed. A stiffer bed cannot remove compliance in the X carriage, adapters or spindle mounting.

Positive mode verification

Confirm all beam/panel states, clamp engagement and head identity before enabling the selected process. Treat these as process interlocks until the electrical architecture is properly reviewed; they are not a certified safety system.

Dryness and contamination

Draining does not dry steel. Inspect the pan, protect the underside of deck panels from condensation, and collect router dust at the head. Remove all wood parts and dust before plasma operation; keep the storage cabinet closed.

Fabrication release

Mock up storage, beam seating and tool sweep at full scale. Then finalize pin/bushing fits, clamp hardware, connections, cabinet openings, plumbing materials and wiring. Prove clean re-docking and loaded deflection with an indicator.